

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering
BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR (Contextualized Construct Representations) text-analysis platform (Python, **FastAPI**, sentence-transformers, React), productionizing a published NLP measurement method into a self-serve tool and owning the model lifecycle from problem definition and data-quality assessment to reproducible, documented outputs.
- Designed the platform's data infrastructure: a YAML model registry (single source of model truth), a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system with deterministic language detection that reports uncertainty rather than guessing; grew the backend suite to **40 hermetic tests**.

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer on a provider-facing **SMART on FHIR** clinical decision support app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization; integrating AI/ML models to estimate medication efficacy and prior-authorization approval likelihood.
- Extending a **Firebase Cloud Functions** backend that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations and aggregation queries; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.

Projects

Refusal Decay – Statistical Inference on LLM Behavior (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Measured how prefilling attacks weaken refusal in Llama-3.1-8B-Instruct: extracted a residual-stream signal via **difference-in-means** (refusal 0.92 → 0.32 at $k=3$), tested it with **causal-intervention** activation patching, and reported effects with **bootstrap 95% confidence intervals** and **McNemar's** significance test.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component and evaluation of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** (54.53 → 57.21) over the KG²RAG baseline on a 4,905-question benchmark with 95% binomial confidence intervals.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a Spark/SQL ETL and analytics pipeline over a **~1.4B-row** NYC TLC dataset, using Spark UI stage metrics to deep-dive shuffle and skew bottlenecks and cut end-to-end runtime **~25%** (17–19 min → 14.3 min) and the tail-latency ratio 2.14x → 1.74x.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, SQL, TypeScript, JavaScript, C++, Bash, PostgreSQL, MongoDB, SQLite

Statistics & Inference: Hypothesis testing (McNemar's), bootstrap / binomial confidence intervals, difference-in-means, causal-intervention analysis, significance testing

ML & Modeling: PyTorch, NumPy, scikit-learn, sentence-transformers, Hugging Face Transformers, JAX

Data & Analytics: Spark / PySpark, Spark UI profiling, ETL optimization, pandas, matplotlib, aggregation queries, data-quality validation

Cloud & DevOps: AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD